Mass points

L3-X1 Level 3 Senior · Geometry · Beyond the textbook

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Masses at the vertices and the lever rule, for every ratio in a figure of cevians and transversals at once, with the link to Menelaus and Ceva and the cases where to switch to them.
By the end you can assign masses from given ratios and read off every ratio on the cevians and sides; split a mass for a line crossing two sides, and find areas from masses; recognise when mass points fail and use Menelaus, Ceva or vectors instead.
Prerequisites: Area, Pythagoras and similar shapes (L2-GE2), Area ratios without lengths (L2-X1).
Contests: SMC, Senior Kangaroo, Maclaurin, AMC 12, AIME. **Time:** about 90 minutes.

1 Lesson

In triangle ABC, the point D lies on BC with $BD : DC = 1 : 2$, and E is the midpoint of CA. The cevians AD and BE meet at P. Find $AP : PD$. Put mass 2 at B and 1 at C, so that D is their balance point. Put mass 1 at A, so that E is the balance point of A and C. Then D carries mass 3, and P is the balance point of A (mass 1) and D (mass 3), so $AP : PD = 3 : 1$. This is three lines of arithmetic. With L2-X1 the problem becomes a labelling problem with two unknown areas, and with coordinates the problem becomes a pair of simultaneous equations.

The idea goes back to Archimedes' law of the lever. Möbius made the idea into a calculus of weighted points in *Der barycentrische Calcul* (1827), and in the form below, with integer masses written at the vertices, the idea is folklore of American contest training under the name *mass points*. Textbooks rarely include the method because everything the method proves also follows from Menelaus and Ceva, or from vectors. Its advantage is speed, since one set of masses gives every ratio on every line of the figure at once. We assume L2-GE2 and L2-X1, and position vectors.

1.1 Masses and the lever rule

A *mass point* mP is a point P with a positive number m attached. We define the sum of two mass points by

$$mP + nQ = (m+n)R, \qquad \text{where } R = \frac{mP+nQ}{m+n}$$

as position vectors. So R is the centre of mass, and its mass is $m+n$.

Theorem 1 (lever rule).
The point R lies on segment PQ with $PR : RQ = n : m$. The heavier mass is nearer to R.
Theorem 2 (grouping). The sum is commutative and associative. Hence the centre of mass of several mass points does not depend on the order in which we group them.

Proofs. $R - P = \frac{n}{m+n}(Q-P)$ and $Q - R = \frac{m}{m+n}(Q-P)$, which gives Theorem 1. For Theorem 2, grouping in any order gives the point $\frac{\sum m_i P_i}{\sum m_i}$ with mass $\sum m_i$. ■

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1.2 Cevians

Theorem 3 (cevians).
Put masses a, b, c at the vertices of triangle ABC, and let G be the centre of mass of all three. Let D, E, F be the centres of $bB + cC$, $cC + aA$, $aA + bB$. Then AD, BE, CF pass through G, and

$$\frac{BD}{DC} = \frac{c}{b}, \quad \frac{CE}{EA} = \frac{a}{c}, \quad \frac{AF}{FB} = \frac{b}{a}, \qquad \frac{AG}{GD} = \frac{b+c}{a}, \quad \frac{BG}{GE} = \frac{c+a}{b}, \quad \frac{CG}{GF} = \frac{a+b}{c}.$$

Conversely, if cevians AD, BE, CF meet at a point P, there are masses with these properties and $G = P$.

Theorem 4 (areas). With G as in Theorem 3, $[GBC] : [GCA] : [GAB] = a : b : c$.

Proofs. By Theorem 2, G is the centre of $aA + (b + c)D$, so G lies on AD, and Theorem 1 gives both ratios on that line. The other two cevians are the same. For the converse, choose b, c with $c : b = BD : DC$, then a with $a : c = CE : EA$. The centre G lies on AD and on BE, so $G = P$, and CF passes through G and meets AB at the centre of $aA + bB$. For Theorem 4, triangles GBC and ABC share the base BC, and their heights are in the ratio $GD : AD = a : (a + b + c)$. ■

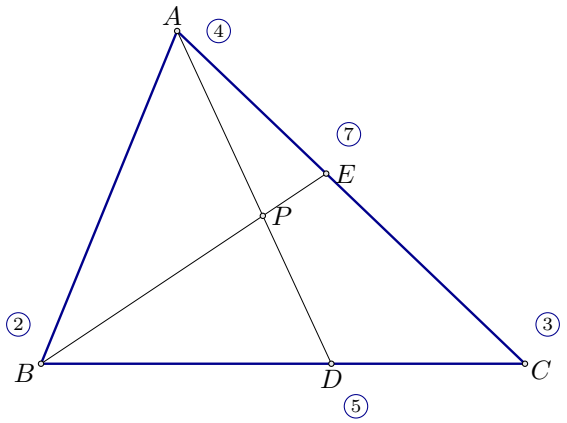
Multiplying the three side ratios gives $\frac{c}{b} \cdot \frac{a}{c} \cdot \frac{b}{a} = 1$, which is Ceva's theorem. In practice, read off the masses from two given ratios, scale them so that the vertex they share gets the same mass, and then every other ratio is a quotient of masses.

Example 1.
In triangle ABC, D lies on BC with $BD : DC = 3 : 2$ and E lies on CA with $CE : EA = 4 : 3$. The cevians AD and BE meet at P. Find $AP : PD$ and $BP : PE$.

Solution.
First the standard way. Line BPE crosses the sides of triangle ADC, meeting AD at P, DC extended at B and CA at E. Menelaus gives

$$\frac{AP}{PD} \cdot \frac{DB}{BC} \cdot \frac{CE}{EA} = 1, \quad \text{so} \quad \frac{AP}{PD} = \frac{5}{3} \cdot \frac{3}{4} = \frac{5}{4}.$$

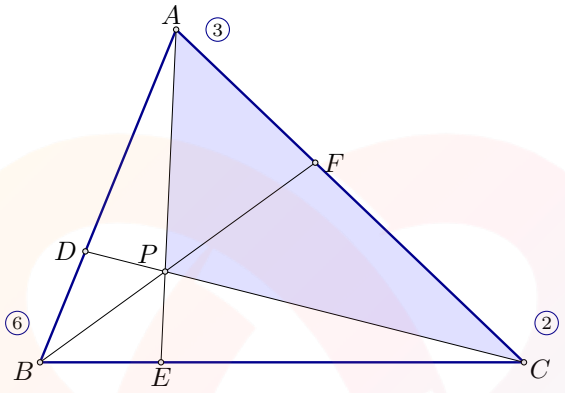
A second Menelaus, in triangle BEC with line APD, gives $BP : PE$. Each needs care with the choice of triangle.
With masses, $BD : DC = 3 : 2$ needs $b : c = 2 : 3$, and $CE : EA = 4 : 3$ needs $c : a = 3 : 4$. Take $a = 4$, $b = 2$, $c = 3$, so D has mass 5 and E has mass 7. By Theorem 3, $AP : PD = 5 : 4$ and $BP : PE = 7 : 2$.



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Example 2.
In triangle ABC of area 44, D lies on AB with $AD : DB = 2 : 1$ and E lies on BC with $BE : EC = 1 : 3$. The lines CD and AE meet at P, and BP meets AC at F. Find $AF : FC$ and the area of triangle APC.

Solution.
$AD : DB = 2 : 1$ needs $a : b = 1 : 2$, and $BE : EC = 1 : 3$ needs $b : c = 3 : 1$. The vertex B appears in both, so we take $b = 6$, which gives $a = 3$ and $c = 2$. Then $AF : FC = c : a = 2 : 3$.
By Theorem 4, $[APC] = \frac{b}{a+b+c} \cdot 44 = \frac{6}{11} \cdot 44 = 24$.
The ratios on the cevians come with no extra work, since $CP : PD = (a + b) : c = 9 : 2$ and $AP : PE = (b + c) : a = 8 : 3$.



1.3 Splitting masses

A line that crosses two sides, rather than passing through a vertex, is not a cevian. Mass points still handle this case if we allow the mass at one vertex to be split into two parts, one for each side.

Theorem 5 (split masses).
A line meets AB at X, AC at Y, and the cevian AD at P. Suppose X is the centre of $a_1A + bB$, Y is the centre of $a_2A + cC$, and D is the centre of $bB + cC$. Then

$$\frac{AP}{PD} = \frac{b + c}{a_1 + a_2}, \qquad \frac{XP}{PY} = \frac{a_2 + c}{a_1 + b}.$$

Proof. Let G be the centre of $(a_1 + a_2)A + bB + cC$. By Theorem 2,

$$(a_1 + a_2)A + bB + cC = (a_1A + bB) + (a_2A + cC) = (a_1 + b)X + (a_2 + c)Y,$$

so G lies on XY, and the same sum equals $(a_1 + a_2)A + (b + c)D$, so G lies on AD. The two lines meet only at P, so $G = P$, and Theorem 1 gives both ratios. ■

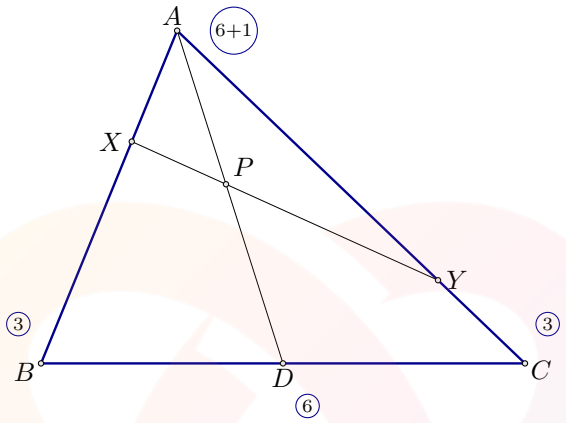
So the mass at A is a_1 when we look along AB, a_2 along AC, and $a_1 + a_2$ along AD.

Example 3.
In triangle ABC, D is the midpoint of BC, X lies on AB with $AX : XB = 1 : 2$, and Y lies on AC with $AY : YC = 3 : 1$. The segment XY meets AD at P. Find $AP : PD$ and $XP : PY$.

Solution.

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First with vectors, taking A as origin and writing $\mathbf{b}$, $\mathbf{c}$ for B, C. Then $X = \frac{1}{3}\mathbf{b}$, $Y = \frac{3}{4}\mathbf{c}$ and $D = \frac{1}{2}(\mathbf{b}+\mathbf{c})$. A point of AD is $\frac{t}{2}(\mathbf{b}+\mathbf{c})$ and a point of XY is $\frac{1-s}{3}\mathbf{b} + \frac{3s}{4}\mathbf{c}$. Comparing coefficients, $\frac{t}{2} = \frac{1-s}{3} = \frac{3s}{4}$, so $s = \frac{4}{13}$ and $t = \frac{6}{13}$. Hence $AP : PD = 6 : 7$ and $XP : PY = 4 : 9$.
With split masses, D is the midpoint, so $b = c$. Take $b = c = 3$. On AB, $AX : XB = 1 : 2$ needs $a_1 : b = 2 : 1$, so $a_1 = 6$. On AC, $AY : YC = 3 : 1$ needs $a_2 : c = 1 : 3$, so $a_2 = 1$. By Theorem 5, $AP : PD = (3+3) : (6+1) = 6 : 7$ and $XP : PY = (1+3) : (6+3) = 4 : 9$.
The vector method needs no new idea and is the right choice in a written proof. The masses give the same numbers in a few seconds, which is what matters in the SMC or the AMC 12.



1.4 Menelaus and Ceva in disguise

Every ratio that mass points produce is a Menelaus or Ceva statement. In Example 1, Menelaus for triangle ADC and line BPE reads, in terms of masses,

$$\frac{AP}{PD} \cdot \frac{DB}{BC} \cdot \frac{CE}{EA} = \frac{b+c}{a} \cdot \frac{c}{b+c} \cdot \frac{a}{c} = 1,$$

and the product of the side ratios in Theorem 3 is Ceva. The masses keep all these statements consistent with each other, so you never have to choose the triangle for Menelaus. You should switch to Menelaus, Ceva or vectors in the following cases.

1. A point lies on the extension of a side. Theorem 1 would need a negative mass. Either use Menelaus, or find another triangle in which every point lies on a side or a cevian (Example 4).
2. A line is parallel to a side. Its intersection with that side does not exist, and the masses at the two ends would have to add up to zero. Use similar triangles.
3. Two lines cross the triangle and neither passes through a vertex. Theorem 5 handles one such line through a cevian. Two transversals meeting each other need vectors or Menelaus.
4. The data are lengths or angles that are not ratios on the lines of the figure. Convert them first, for example with the angle bisector theorem.
5. A written proof, as in the Maclaurin or BMO1. State the vector form of Theorems 1 to 5, or give a Menelaus argument. Masses written at the vertices with no explanation are hard for a marker to check.

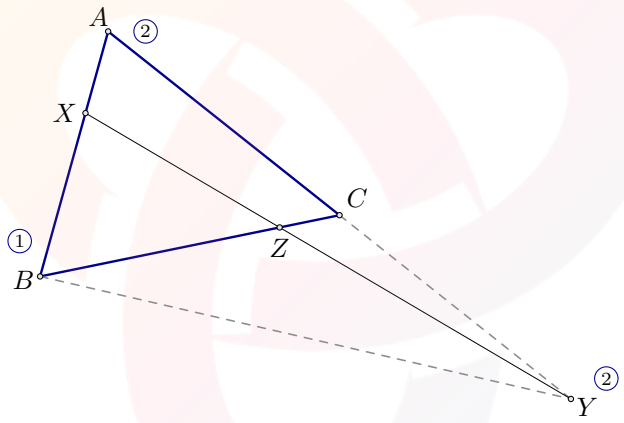
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Example 4.
In triangle ABC, X lies on AB with $AX : XB = 1 : 2$, and Y lies on ray AC beyond C with $CY = AC$. The line XY meets BC at Z. Find $BZ : ZC$ and $XZ : ZY$.

Solution.
The point Y is outside triangle ABC, so masses on ABC do not apply. Look at triangle ABY instead. In this triangle, C lies on side AY, X lies on side AB, and Z is the intersection of the cevians BC and YX. Now C is the midpoint of AY, so $a = y$, and $AX : XB = 1 : 2$ needs $a : b = 2 : 1$. Take $a = y = 2$ and $b = 1$. Then C has mass 4 and X has mass 3. By Theorem 3, $BZ : ZC = 4 : 1$ and $YZ : ZX = 3 : 2$, so $XZ : ZY = 2 : 3$.
Check with Menelaus for triangle ABC and the line XZY:

$$\frac{AX}{XB} \cdot \frac{BZ}{ZC} \cdot \frac{CY}{YA} = \frac{1}{2} \cdot 4 \cdot \frac{1}{2} = 1.$$

Choosing the triangle is the hard step. Look for a vertex whose two lines contain all the other points that lie on sides, which here is A, with lines AB and AY. The far points B and Y on those lines are the other two vertices.



Common traps.

- Putting the masses the wrong way round. If $BD : DC = 3 : 2$ then D is nearer C, so C is heavier and $b : c = 2 : 3$.
- Not rescaling. A vertex that appears in two given ratios must get the same mass in both. Scale by the least common multiple.
- With split masses, using $a_1 + a_2$ on AB, or a_1 on AD. Each line sees only the masses whose centre lies on that line.
- Reading lengths from masses. Masses give ratios on a line through two massed points, not ratios between different lines, and not lengths unless one length on that line is known.
- Using Theorem 4 for a triangle whose vertex is not the centre of mass G. Other areas need L2-X1.

1.5 How to spot this in a contest

1. Two ratios on two sides of a triangle, and a ratio on a cevian is asked for, or the reverse.

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2. Three concurrent cevians with some ratios given and an area asked for. Use Theorems 3 and 4.
3. A segment joining points on two sides crossing a cevian, often a median. Split the mass at the common vertex.
4. A point on an extended side. Change the triangle, or use Menelaus.
5. An angle bisector or a midpoint in the statement. These are ratios in disguise and give masses directly.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ In triangle ABC, D lies on BC with $BD : DC = 3 : 1$ and E lies on CA with $AE : EC = 2 : 3$. The cevians AD and BE meet at P. Find $AP : PD$.

Problem 2 ⋆⋆ The cevians AD, BE, CF of triangle ABC meet at P. The triangle has area 60, $BD : DC = 2 : 3$ and $CE : EA = 1 : 2$. Find the area of triangle PAB.

Problem 3 ⋆⋆ In triangle ABC, D lies on BC with $BD : DC = 1 : 2$, X lies on AB with $AX : XB = 1 : 3$, and Y lies on AC with $AY : YC = 2 : 1$. The segment XY meets AD at P. Find $AP : PD$.

Problem 4 ⋆⋆⋆ The cevians AD, BE, CF of triangle ABC meet at P, with $AP : PD = 3 : 1$ and $BP : PE = 2 : 1$. Find $CP : PF$.

Problem 5 ⋆⋆⋆ In triangle ABC, X lies on AB with $AX : XB = 2 : 1$, and Y lies on ray BC beyond C with $BC : CY = 2 : 1$. The line XY meets AC at Z. Find $AZ : ZC$.

Problem 6 ⋆⋆⋆⋆ In triangle ABC, D lies on BC with $BD : DC = 2 : 3$, and E is the midpoint of CA. The cevians AD and BE meet at P, and CP meets AB at F. The segment EF meets AD at Q. Find $AQ : QP$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **ARML 1989 Team Problem 4** ⋆
   By the angle bisector theorem, masses proportional to the side lengths do the job.
2. **AHSME 1971 Problem 26** ⋆
   Masses at A and C from AF:FC, then B from the midpoint G.
3. **AMC 10B 2004 Problem 20** ⋆
   Work backwards, choosing masses that give the two ratios on the cevians.
4. **ARML 1992 Individual Problem 8** ⋆⋆
   The bisector AT is a cevian and DE crosses two sides, so split the mass at A (Theorem 5).

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5. **Purple Comet 2010 High School Problem 16** ⋆⋆
   Only the ratios on AD matter, not the side lengths. Use one set of masses for E and another for F.

6. **AMC 12B 2025 Problem 10** ⋆⋆
   Find where the foot of the altitude divides the hypotenuse, then use masses.

7. **AIME I 2009 Problem 4** ⋆⋆
   Split the mass at A between AB and AD. The diagonal AC passes through the centre of B and D.

8. **AIME II 2011 Problem 4** ⋆⋆
   Masses at B and C from the bisector, then the midpoint M of AD fixes A.

9. **AIME I 2009 Problem 5** ⋆⋆⋆
   Use the midpoint of PM to find a parallelogram, then masses on BK.

10. **AIME 1988 Problem 12** ⋆⋆⋆
    Write each cevian ratio as a quotient of masses, or of areas, and add.

11. **APMO 1989 Problem 3** ⋆⋆⋆
    Each D and each E is the meeting point of two cevians of the original triangle. Find its masses, then use Theorem 4 or L2-X1 for the areas.

12. **AIME 1992 Problem 14** ⋆⋆⋆
    Express each ratio AO/OA′ in terms of a, b, c and compare the sum with the product.

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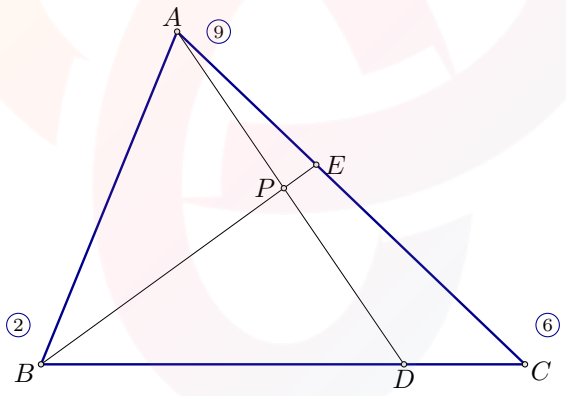
4 Hints

1. Which end of each side is heavier? Then make the mass at C agree.
2. Find the masses, then use Theorem 4.
3. Split the mass at A into one part for AB and one for AC.
4. Write the two given ratios in terms of the masses a, b, c and solve.
5. Y is outside triangle ABC. Find a triangle in which C, X and Z are all on sides or cevians.
6. E and F both use the mass at A. Give A its mass twice, once for E and once for F, and apply Theorem 5 to the line EF.

5 Solutions

Problem 1

In triangle ABC, D lies on BC with $BD : DC = 3 : 1$ and E lies on CA with $AE : EC = 2 : 3$. The cevians AD and BE meet at P. Find $AP : PD$.



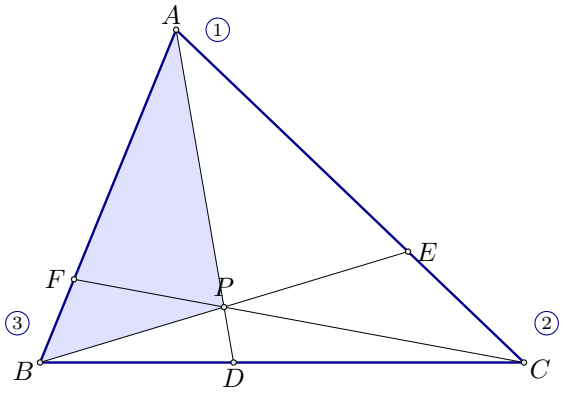
D is nearer C, so $b : c = 1 : 3$. E is nearer A, with $a : c = 3 : 2$. The common vertex is C. Take $c = 6$, so $b = 2$ and $a = 9$. Then D has mass 8, and by Theorem 3, $AP : PD = (b + c) : a = 8 : 9$.

Answer: $8 : 9$

Problem 2

The cevians AD, BE, CF of triangle ABC meet at P. The triangle has area 60, $BD : DC = 2 : 3$ and $CE : EA = 1 : 2$. Find the area of triangle PAB.

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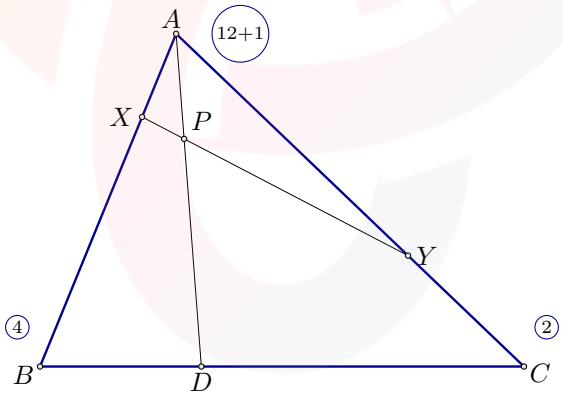


$BD : DC = 2 : 3$ needs $b : c = 3 : 2$, and $CE : EA = 1 : 2$ needs $c : a = 2 : 1$. So $a = 1$, $b = 3$, $c = 2$, and the point P is the centre of mass. By Theorem 4, $[PAB] = \frac{c}{a+b+c} \cdot 60 = \frac{2}{6} \cdot 60 = 20$. (The masses also give $AF : FB = b : a = 3 : 1$, and Ceva checks that $\frac{2}{3} \cdot \frac{1}{2} \cdot 3 = 1$.)

Answer: 20

Problem 3

In triangle ABC, D lies on BC with $BD : DC = 1 : 2$, X lies on AB with $AX : XB = 1 : 3$, and Y lies on AC with $AY : YC = 2 : 1$. The segment XY meets AD at P. Find $AP : PD$.



$BD : DC = 1 : 2$ needs $b : c = 2 : 1$. Take $b = 4$, $c = 2$. On AB, $AX : XB = 1 : 3$ needs $a_1 : b = 3 : 1$, so $a_1 = 12$. On AC, $AY : YC = 2 : 1$ needs $a_2 : c = 1 : 2$, so $a_2 = 1$. By Theorem 5,

$$\frac{AP}{PD} = \frac{b+c}{a_1+a_2} = \frac{6}{13}.$$

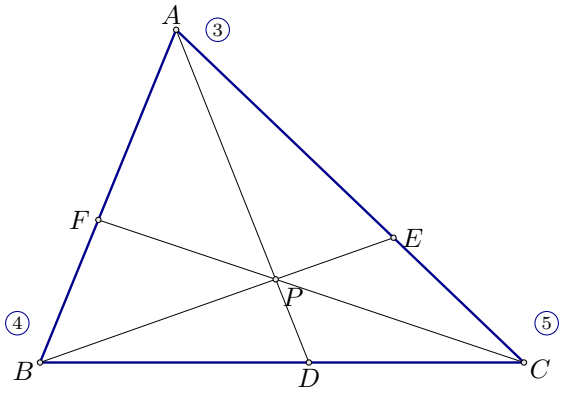
(Also $XP : PY = (a_2 + c) : (a_1 + b) = 3 : 16$, so P is close to X, as the figure shows.)

Answer: $6 : 13$

Problem 4

The cevians AD, BE, CF of triangle ABC meet at P, with $AP : PD = 3 : 1$ and $BP : PE = 2 : 1$. Find $CP : PF$.

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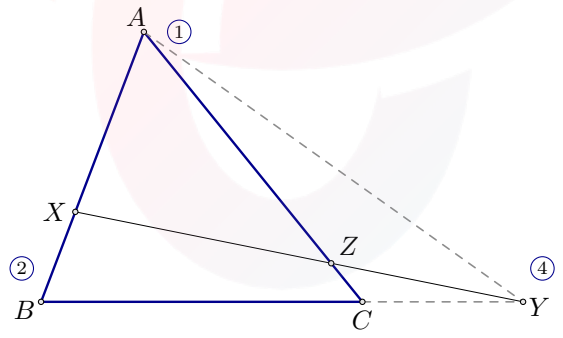
By the converse part of Theorem 3 there are masses a, b, c with centre P, and then $\frac{b+c}{a} = 3$ and $\frac{a+c}{b} = 2$. Masses can be scaled, so take $a = 3$. Then $b + c = 9$ and $3 + c = 2b$, which give $b = 4$ and $c = 5$. Hence $CP : PF = (a + b) : c = 7 : 5$.

Without masses this is a system of Menelaus equations. With masses the system is two linear equations.

Answer: $7 : 5$

Problem 5

In triangle ABC, X lies on AB with $AX : XB = 2 : 1$, and Y lies on ray BC beyond C with $BC : CY = 2 : 1$. The line XY meets AC at Z. Find $AZ : ZC$.



Work in triangle ABY. There C lies on side BY with $BC : CY = 2 : 1$, so $b : y = 1 : 2$, and X lies on side AB with $AX : XB = 2 : 1$, so $a : b = 1 : 2$. Take $a = 1$, $b = 2$, $y = 4$. The point Z is the intersection of the cevians AC and YX, and C has mass $b + y = 6$, so $AZ : ZC = 6 : 1$.

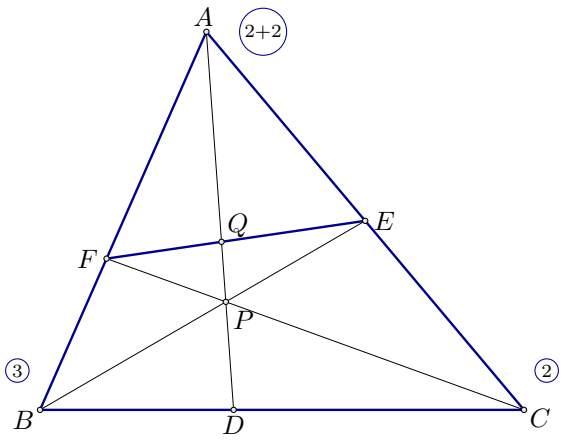
Check with Menelaus for triangle ABC and the line XZY. Then $\frac{AX}{XB} \cdot \frac{BY}{YC} \cdot \frac{CZ}{ZA} = 2 \cdot 3 \cdot \frac{1}{6} = 1$.

Answer: $6 : 1$

Problem 6

In triangle ABC, D lies on BC with $BD : DC = 2 : 3$, and E is the midpoint of CA. The cevians AD and BE meet at P, and CP meets AB at F. The segment EF meets AD at Q. Find $AQ : QP$.

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$BD : DC = 2 : 3$ needs $b : c = 3 : 2$, and E is the midpoint of CA, so $a = c$. Take $a = 2$, $b = 3$, $c = 2$. Then P is the centre of mass, F is the centre of $2A + 3B$, and $AP : PD = (b + c) : a = 5 : 2$.

For Q, split masses on line EF, where F is the centre of $2A + 3B$ and E is the centre of $2A + 2C$, so by Theorem 5 with $a_1 = a_2 = 2$, the point Q is the centre of $4A + 3B + 2C$ and $AQ : QD = (3 + 2) : 4 = 5 : 4$.

So $AQ = \frac{5}{9}AD$ and $AP = \frac{5}{7}AD$, giving $QP = \left(\frac{5}{7} - \frac{5}{9}\right) AD = \frac{10}{63}AD$ and $AQ : QP = \frac{35}{63} : \frac{10}{63} = 7 : 2$.

The same computation with general masses gives $AQ : QP = (a + b + c) : a$, and $AD : DP = (a + b + c) : a$ as well, so Q and D divide AP internally and externally in the same ratio. This is a harmonic range, which you will meet at Level 4.

Answer: $7 : 2$

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